

Homework 2: Data and Distributions

Fitting power-law, exponential, uniform and normal models

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CS 6330: Introduction to Data Science

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Data. `airport_routes.csv` (3409 airports, Feb 2017), `movie_votes.csv` (4392 TMDb average ratings), `DoB.txt` (460,718 IMDb birth dates).

Tooling. Python 3 / pandas / numpy / scipy / matplotlib; all estimates computed programmatically, none typed by hand.

Problem 1: Fit four distributions to two datasets

(90 pts)

Dataset A: airport routes

- (a) Power law: α
- (b) Exponential: λ
- (c) Uniform: $[a, b]$
- (d) Normal: μ and σ

Distribution of the data

Which model fits, and why

Dataset B: movie votes

- (a) Power law: α
- (b) Exponential: λ
- (c) Uniform: $[a, b]$
- (d) Normal: μ and σ

Distribution of the data

Which model fits, and why

Problem 2: Age, first digit and last digit

(30 pts)

Distribution of age

Distribution of the first digit

Distribution of the last digit

Discussion: which digits are uniform, and why